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Arogya Sathi: proactive "Virtual Nurse."

A Sovereign, Agentic Digital Health Ecosystem for Urban India

1. Problem Statement

can keep this in motivation but not proble statement select problem statement properly and keep it in bullet points

The NCD Epidemic

Urban India faces a **26.4% diabetes prevalence** (ICMR, 2023).

Generic step-trackers fail to provide the medical-grade monitoring required for this specific demographic. The sheer volume of undiagnosed conditions creates a massive socio-economic burden.

The ”Pollution-Blind” Gap

Standard health algorithms ignore environmental toxicity, dangerously suggesting "outdoor runs" during smog events.

They fail to triage the **200,000+ emergency respiratory cases** directly linked to poor air quality in cities like Delhi.

2. Motivation & Need

in motivation add my fathers problem as well that he has a high colestrol and he just keeps asking us things and for that i thought of something personalized and easy for him

Macroeconomic Threat

NCDs project a **$4.58 Trillion** economic loss for India by 2030 (WEF).

Preventative digital healthcare is no longer just a luxury; it is an absolute economic imperative to sustain national growth and workforce productivity.

The Linguistic Divide

90% of new internet users in India consume vernacular content, yet over 90% of health apps are English-only.

A native-language "Virtual Nurse" is essential to bridge the digital health literacy gap and democratize preventative care.

3. Project Objectives

Agentic Triage

Deploy an AI "Virtual Nurse" to assess symptoms based on ICMR guidelines, strictly constrained to avoid medical diagnosis.

Hyper-Local Logic

Integrate live APIs (WAQI, OpenWeather) to dynamically alter health advice based on localized smog alerts and weather.

Data Sovereignty

Architect a 100% Indian-hosted, DPDP Act (2023) compliant ecosystem where health records never cross national borders.

4. Project Scope

can remove out of scope and highlighted thing from in scope with that add proper in scope things and better in bullet points

✓ In-Scope

- → 12-Week Capstone timeline execution.
- → Cross-platform app (Flutter) with offline caching.
- → FastAPI AI microservices integration.
- → ICMR-grounded RAG pipeline (BioMistral).
- → OCR extraction for physical lab reports.

✗ Out-of-Scope

- → **Medical Diagnoses or Prescriptions.**
- → Hardware wearable integration (Apple Watch, etc).
- → Live ABDM production deployment (sandbox only).
- → Treatment of acute medical emergencies.

5. Literature Review

add a proper things about litrature review form our current litrature reiwe with proper bullet points (you can use more that one slide for one topic if needed like for this or anything)

Foundation built exclusively on peer-reviewed clinical and technical research.

Epidemiological Basis

- → **ICMR-INDIAB Study (2023):** Establishes the massive 26.4% urban diabetes baseline, justifying the focus on NCDs in metropolitan areas.
- → **AQLI Report (2024):** Quantifies the 11.9-year life expectancy loss in Delhi, providing absolute validation for our AQI-aware triage system.

Technical & Legal Basis

- → **Labrak et al. (2024) - BioMistral:** Proves the feasibility of quantized, medical-specific open-source models functioning without external cloud APIs.
- → **DPDP Act (2023):** Acts as the legal blueprint for our sovereign database schema, enforcing strict local data residency.

6. Proposed Methodology

hey proposed methodology should include some architectural digram (i will attach kind of one) with that it should include flow that how system will work instead of this (look here you will have much more info like digram and flow so in such places to keep ppt clean and informative dont cut out the info just take that topic on next slide)

Replacing static decision trees with an autonomous Agentic Workflow Execution.

1. Intercept

Fetch live GPS and WAQI (Air Quality) environmental data continuously in the background.

2. Retrieve

Query ICMR guidelines via ChromaDB to ensure protocols are medically grounded.

3. Reason

BioMistral-7B LLM correlates the reported symptoms directly with the environmental context.

4. Filter

A strict heuristic block scans output to remove any diagnostic or prescriptive language before delivery.

7. Tools & Technologies

tools and tech should be listed properly and resonabaly with proper bullet points and all the things

Client & Backend

Flutter: Offline-first UI via local SQLite database.

FastAPI & PostgreSQL: High-performance asynchronous backend for orchestrating external data APIs securely.

AI & ML Stack

BioMistral-7B: Open-source, medically fine-tuned LLM.

Sarvam AI & Bhashini: Domestic models utilized exclusively for vernacular speech-to-text and translation.

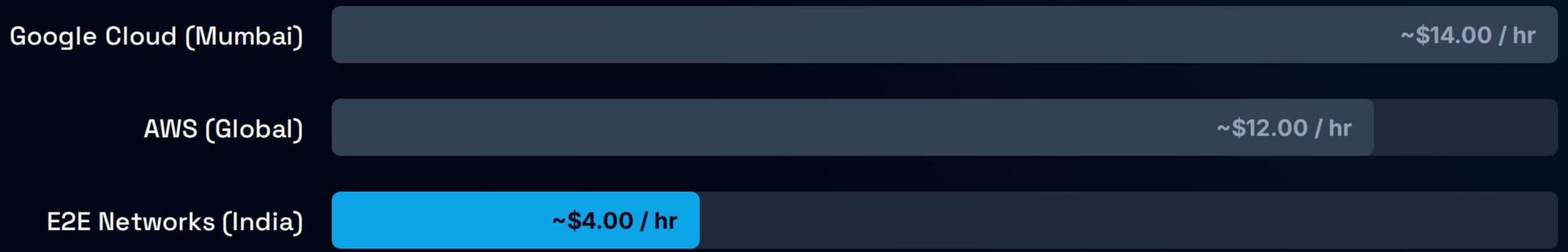
Sovereign Cloud

E2E Networks: Indian-based data centers providing NVIDIA GPUs to ensure strict DPDP Act compliance and lower costs.

8. Feasibility of Implementation

include more and proper points in this dont just add about processing and all add about every fisibility and implementation and make it detailed and proper

High-parameter AI is cost-prohibitive. We resolve this via **Sovereign Cloud Arbitrage**.



> *By utilizing domestic providers (E2E Networks) and 4-bit model quantization, we reduce H100 GPU inference costs by **~70%**, ensuring financial viability for an academic capstone and real-world scalability.*

9. Expected Outcomes

Technical Deliverables

- → **Functional MVP App:** Android delivery with vernacular voice onboarding and offline caching capability.
- → **Context-Aware Triage:** Backend successfully correlating symptoms with live AQI > 300 to dynamically alter advice.
- → **Sovereign Architecture PoC:** Validation of zero-knowledge local inference without relying on OpenAI/Google.

Academic & Public Impact

- → **Research Output:** Comprehensive architecture report, API schemas, and accuracy benchmarking for the RAG pipeline.
- → **Reduced False Alarms:** Lowering unnecessary clinical visits by attributing symptoms to pollution instead of infection.
- → **Digital Inclusion:** Bridging the health literacy gap for non-English speakers.

10. Project Timeline (12 Weeks)

design a gantt chart properly and relastic like professional gantt chart

TASK DESCRIPTION	MONTH 1 (W1-W4)	MONTH 2 (W5-W8)	MONTH 3 (W9-W12)
1. DB & API Architecture	• PostgreSQL, API schemas		
2. Frontend & Offline SQLite	• Flutter UI/UX	• Offline sync dev	
3. Backend & Microservices		• FastAPI, Context Service	
4. Voice & Env Integration		• WAQI, Bhashini Hooks	
5. LLM & RAG Pipeline			• BioMistral, ChromaDB
6. Testing & Documentation			• Safety Validation, Report

11 & 12. References & Q&A

Primary Citations

- → ICMR-INDIAB Study (2023). "Metabolic non-communicable disease health report of India." The Lancet Diabetes & Endocrinology.
- → AQLI Report (2024). "India Fact Sheet - Air Quality Life Index." University of Chicago.
- → Labrak et al. (2024). "BioMistral: Open-Source Models for Medical Domains."
- → DPDP Act 2023, Ministry of Law & Justice, Govt. of India.

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Q&A

Thank you for your time.

We are open for questions.